

Question 1 (15 marks)

Marks

(a) Find $\int_0^1 x(5x^2 - 2)^4 dx$ 3

(b) Find $\int \cot x dx$ 2

(c) Find $\int \frac{1}{x(x^2 - 1)} dx$ 3

(d) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{dx}{1 - \cos x}$ 4

(e) Evaluate $\int_e^{e^2} \log_e x dx$ 3

Question 2 (15 marks) Start a new page

Marks

(a) Let $z = 3 + 2i$

(i) Find $\bar{z}$ 1

(ii) Find $\frac{1}{z}$ in the form $x + iy$ 2

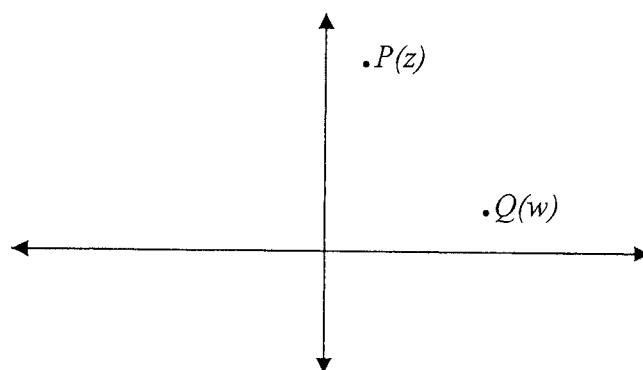
(iii) Find z^{-2} in the form $x + iy$ 1

(b) (i) Express $1 - \sqrt{3}i$ in modulus-argument form. 2

(ii) Find $(1 - \sqrt{3}i)^5$ in the form $x + iy$ 2

(c) Sketch the region in the complex plane where the inequalities $z \leq 2$ and $|\arg z| \leq \frac{\pi}{4}$ hold simultaneously. 2

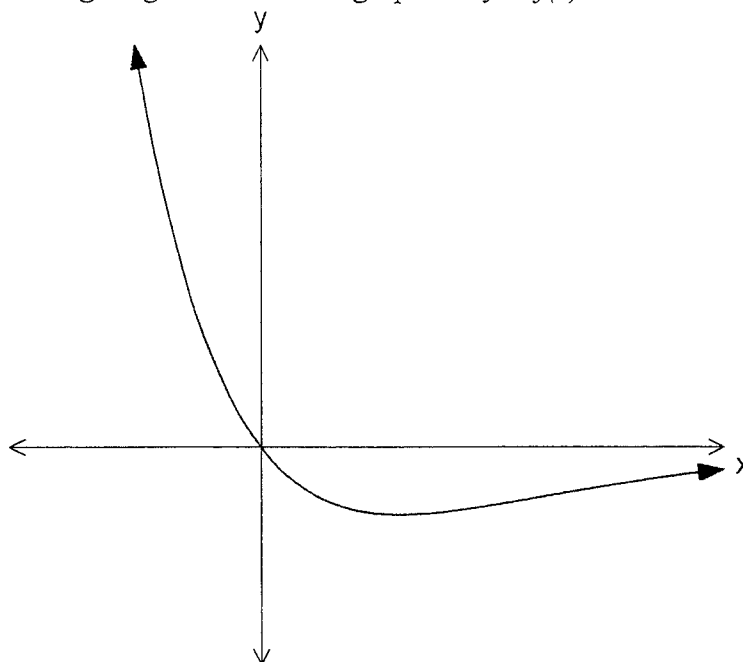
(d) The points P and Q on the Argand diagram represent the complex numbers z and w respectively



Copy the diagram and mark on it the following points:

- | | | |
|-------|------------------------------------|---|
| (i) | The point A representing $-z$ | 1 |
| (ii) | The point B representing $2w$ | 1 |
| (iii) | The point S representing $\bar{z}$ | 1 |
| (iv) | The point T representing iw | 1 |
| (v) | The point U representing $z + w$ | 1 |

(a) The following diagram shows the graph of $y = f(x)$



Draw separate one-third page sketches of the graphs of the following:

- | | | |
|-------|----------------------|---|
| (i) | $y = f(x) $ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y = e^{f(x)}$ | 2 |
| (iv) | $y = f(x)$ | 2 |

(b) Find the coordinates of the points where the tangent to the curve $x^2 + xy + y^2 = 12$ is horizontal 3

(c) The zeros of $2x^3 - 3x^2 + 4x - 1$ are α, β and γ
Find a cubic polynomial with integer coefficients whose zeros are

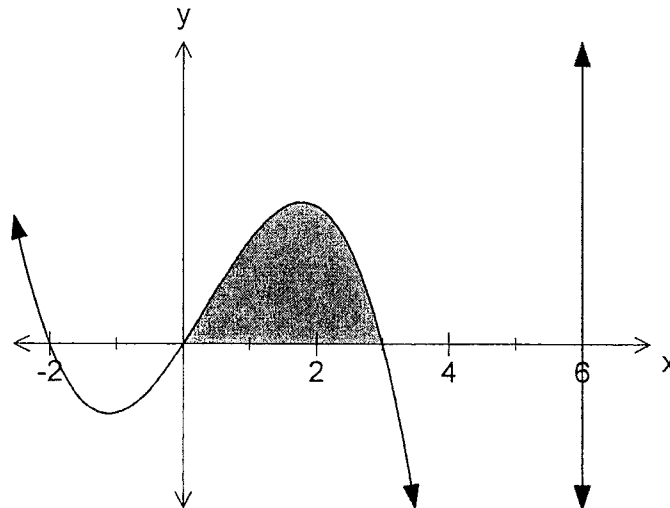
- | | | |
|------|------------------------------------|---|
| (i) | $2\alpha, 2\beta$ and 2γ | 2 |
| (ii) | α^2, β^2 and γ^2 | 3 |

Question 4 (15 marks) Start a new page

Marks

- (a) The region shaded in the diagram is bounded by the x -axis and the curve $y = 6x + x^2 - x^3$

4



The shaded region is rotated about the line $x = 6$.
Find the volume generated.

- (b) (i) Show that the equation of the tangent at the point (x_1, y_1) on the

2

ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = 1$

- (ii) Find the equation of the tangent that passes through the point

1

$(1, \frac{3\sqrt{3}}{2})$ on the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$

- (iii) Find the equation of the tangent parallel to the one in (ii)

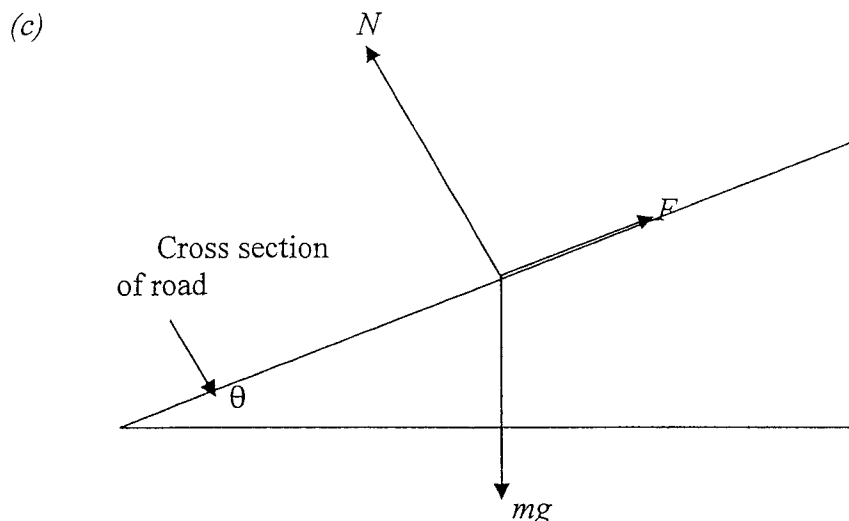
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- (iv) Find the equation of the chord joining the points of contact of the tangents in (ii) and (iii)

1

Question 4 (continued)

Marks

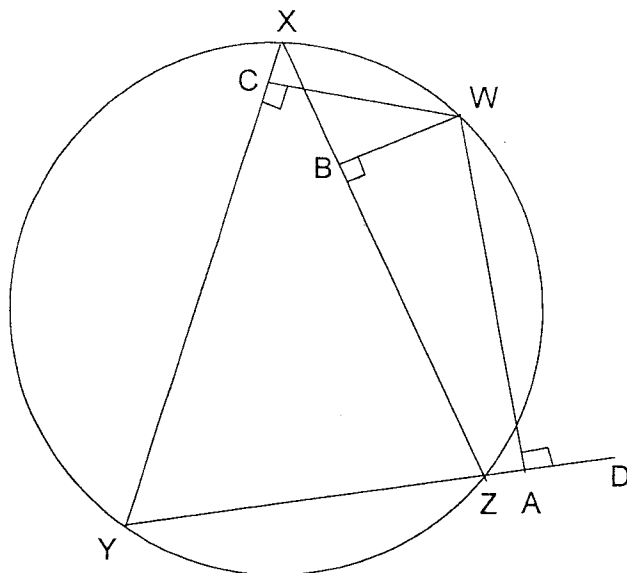


A road contains a bend that is part of a circle radius r . At the bend, the road is banked at an angle θ to the horizontal. A car travels around the bend at constant speed v . Assume that the car is represented by a point of mass m , and that the forces acting on the car are the gravitational force mg , a sideways friction force F (acting up the road as drawn) and a normal reaction N to the road.

- (i) By resolving the horizontal and vertical components of force, find expressions for $N \cos \theta$ and $N \sin \theta$ 3

- (ii) Show that $N = \frac{m(v^2 + gr \cot \theta)}{r} \sin \theta$ 2

(a)



In the diagram W, X, Y and Z are concyclic, and the points A, B, C are the perpendiculars from W to YZ produced, ZX and XY respectively.

- | | | |
|-------|--|---|
| (i) | Show that $\angle WBA = \angle WZA$ | 2 |
| (ii) | Show that $\angle WBC + \angle WXC = 180^\circ$ | 2 |
| (iii) | Deduce that the points A, B and C lie in the same straight line. | 2 |

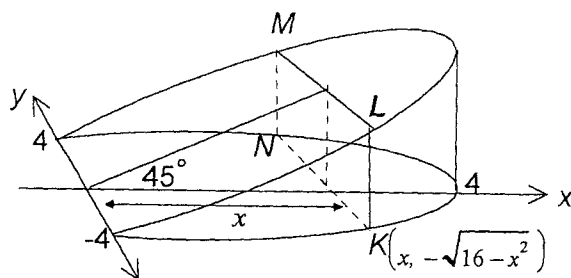
(b) For each integer $n \geq 0$, let $I_n = \int_0^1 x^n e^x dx$

- | | | |
|------|---|---|
| (i) | Show that for $n \geq 1$, $I_n = e - nI_{n-1}$ | 2 |
| (ii) | Hence, or otherwise, calculate I_4 | 2 |

Question 5 (continued)

Marks

- (c) The base of a right cylinder is the circle in the xy -plane with centre O and radius 4. A wedge is obtained by cutting this cylinder with the plane through the y -axis at 45° to the xy -plane, as shown in the diagram.



A rectangular slice $KLMN$ is taken perpendicular to the base of the wedge at a distance x from the y -axis.

- | | | |
|------|--|---|
| (i) | Show that the area of $KLMN$ is given by $x\sqrt{64-4x^2}$ | 2 |
| (ii) | Find the volume of the wedge. | 3 |

Question 6 (15 marks) Start a new page

Marks

(a) Let w be the complex number satisfying $w^3 = 1$ and $\text{Im}(w) > 0$

- | | | |
|-------|---|---|
| (i) | Show that $1 + w + w^2 = 0$ | 2 |
| (ii) | Simplify $w^4 + w^6 + w^8$ | 2 |
| (iii) | Show that $\frac{1}{w^2}$ is a zero of $P(x) = x^4 + 3x^3 + 2x^2 + x - 1$ | 2 |

(b) (i)	Show that $\int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = \frac{\pi}{2}$	2
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(ii)	By making the substitution $x = \pi - u$,	3
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find $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$

(c) (i)	Show that the equation of the tangent at the point $P(ct, \frac{c}{t})$ on the hyperbola $xy = c^2$ is $x + t^2 y = 2ct$	2
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(ii)	Find the equation of the locus of the mid point PG if G is the x intercept of the tangent in (i)	2
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- (a) The curves $y = \sin x$ and $y = \cot x$ intersect at a point A whose x -coordinate is a

(i) Show that $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$ 1

(ii) Show that the curves intersect at right angles at A 3

(iii) Show that $\operatorname{cosec}^2 a = \frac{1+\sqrt{5}}{2}$ 2

- (b) The force of attraction between the earth and a communications

satellite in circular orbit around it is given by $F = \frac{mgR^2}{x^2}$ where

x is the distance of the satellite from the earth's centre, m is the mass of the satellite, g is gravity and R is the radius of the earth. A 300kg satellite is orbiting the earth at 3000m above the surface of the earth.

If $R = 6400\text{km}$ and $g = 10\text{ms}^{-2}$ find

- (i) The velocity of the satellite correct to one significant figure 2
 (ii) The period of the satellite correct to the nearest minute 2
 (iii) F 1

- (c) (i) Differentiate $\sin^{-1} x - \sqrt{1-x^2}$ 2
 (ii) Hence show that

$$\int_0^a \frac{\sqrt{1+x}}{\sqrt{1-x}} dx = \sin^{-1} a + 1 - \sqrt{1-a^2} \text{ for } 0 < a < 1$$
 2

Question 8 (15 marks) Start a new page

Marks

- (a) (i) Show that $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$ 1
- (ii) Show that $\sin \frac{\theta}{2} (1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) = \sin \frac{7\theta}{2}$. 2
- (iii) Show that if $\theta = \frac{2\pi}{7}$, then $1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta = 0$. 2
- (iv) By writing $1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta$ in terms of $\cos \theta$ prove that $\cos \frac{2\pi}{7}$ is a solution of $8x^3 + 4x^2 - 4x - 1 = 0$ 2

(b) Consider the function $f(x) = e^x \left(1 - \frac{x}{8}\right)^8$

- (i) Find the turning points of the graph of $y = f(x)$. 2
- (ii) Sketch the curve $y = f(x)$ and label the turning points and any asymptotes. 2
- (iii) From your graph deduce that $e^x \leq \left(1 - \frac{x}{8}\right)^{-8}$ for $x < 8$. 2
- (iv) Using (iii), show that $\left(\frac{9}{8}\right)^8 \leq e \leq \left(\frac{8}{7}\right)^8$ 2

--- End of Exam ---

(a)

$$\int_0^1 x(5x^2 - 2)^4 dx$$

$$\text{let } u = 5x^2 - 2, \frac{du}{dx} = 10x \Rightarrow \frac{du}{10} = x dx$$

$$\text{when } x = 0, u = -2 \text{ \& } x = 1, u = 3$$

$$\begin{aligned} \therefore I &= \int_{-2}^3 \frac{u^4}{10} du = \left[\frac{u^5}{50} \right]_{-2}^3 = \frac{3^5 - (-2)^5}{50} \\ &= \frac{243 + 32}{50} = \frac{11}{2} \text{ or } 5.5 \end{aligned}$$

(b)

$$\int \cot x dx = \int \frac{\cos x}{\sin x} dx = \ln(\sin x) + C$$

(c)

$$\int \frac{1}{x(x^2 - 1)} dx$$

$$\frac{1}{x(x^2 - 1)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x-1}$$

$$1 = A(x^2 - 1) + Bx(x-1) + Cx(x+1)$$

$$x = 1 \Rightarrow 1 = 2C \quad C = \frac{1}{2}$$

$$x = 0 \Rightarrow 1 = -A \quad A = -1$$

$$x = -1 \Rightarrow 1 = 2B \quad B = \frac{1}{2}$$

$$I = \int \left(\frac{-1}{x} + \frac{1}{2} \left(\frac{1}{x+1} + \frac{1}{x-1} \right) \right) dx$$

$$\therefore I = -\ln x + \frac{1}{2} (\ln(x+1) + \ln(x-1)) + C$$

(d)

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{dx}{1 - \cos x}$$

$$t = \tan \frac{x}{2}, dx = \frac{2dt}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}$$

$$\therefore I = \int_{\tan(\frac{\pi}{6})}^{\tan(\frac{\pi}{2})} \frac{2dt}{1+t^2} = \int_{\frac{1}{\sqrt{3}}}^1 \frac{2dt}{1+t^2}$$

$$= \int_{\frac{1}{\sqrt{3}}}^1 \frac{2dt}{1+t^2} = \int_{\frac{1}{\sqrt{3}}}^1 \frac{2}{1+t^2} dt = \left[\frac{2 \cdot \frac{1}{2} \ln \left(\frac{1+t}{1-t} \right) \right]_{\frac{1}{\sqrt{3}}}^1$$

$$= \left(\frac{1}{1} \right) - \left(\frac{1}{\frac{1}{\sqrt{3}}} \right) = \sqrt{3} - 1$$

(e)

$$\int_e^{e^2} \log_e x dx$$

$$U = \log_e x, U' = 1$$

$$U' = \frac{1}{x}, V = x$$

$$I = x \log_e x - \int dx$$

$$= x \log_e x - x$$

$$\int_e^{e^2} \log_e x dx = [x \log_e x - x]_e^{e^2}$$

$$= e^2 \log_e e^2 - e^2 - (e \log_e e - e)$$

$$= 2e^2 - e^2 - e + e = e^2$$

(a)(i)

$$z = 3 + 2i \quad \bar{z} = 3 - 2i$$

(a)(ii)

$$\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{3-2i}{9-4i^2} = \frac{3}{13} - \frac{2}{13}i$$

(a)(iii)

$$z^{-2} = \left(\frac{1}{z} \right)^2 = \left(\frac{3-2i}{13} \right)^2 = \frac{9-12i+4i^2}{169} = \frac{5}{169} - \frac{12}{169}i$$

(b)(i)

$$1 - \sqrt{3}i \quad \theta = -\tan^{-1} \left(\frac{\sqrt{3}}{1} \right) = -\frac{\pi}{3}$$

$$r^2 = 1^2 + (\sqrt{3})^2 \Rightarrow r = 2$$

$$\therefore 1 - \sqrt{3}i = 2 \operatorname{cis} \left(-\frac{\pi}{3} \right)$$

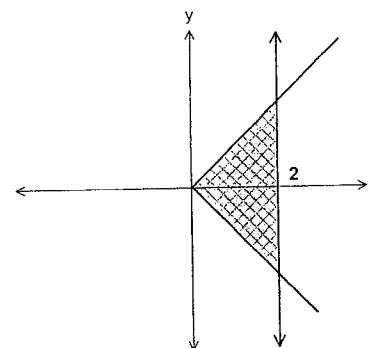
(b)(ii)

$$(1 - \sqrt{3}i)^5 = \left[2 \operatorname{cis} \left(-\frac{\pi}{3} \right) \right]^5 = 32 \operatorname{cis} \left(-\frac{5\pi}{3} \right)$$

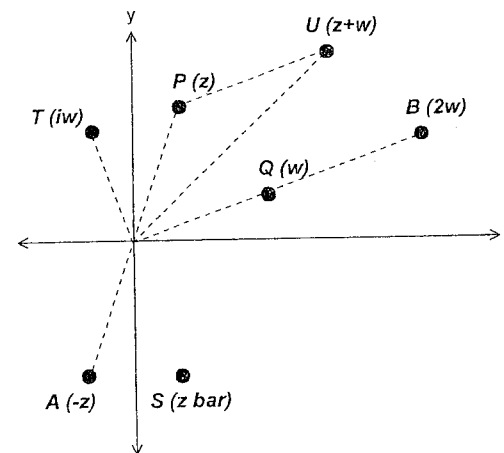
$$= 32 \operatorname{cis} \left(\frac{\pi}{3} \right) = 32 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = 16 + 16\sqrt{3}i$$

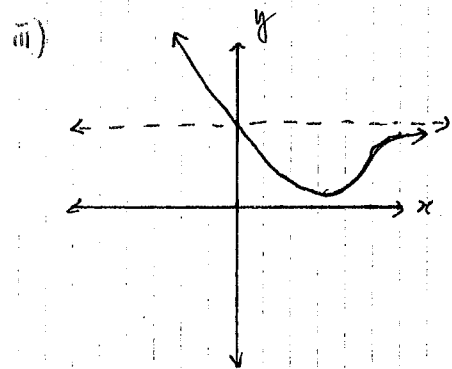
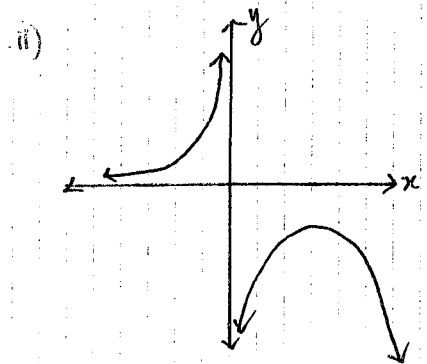
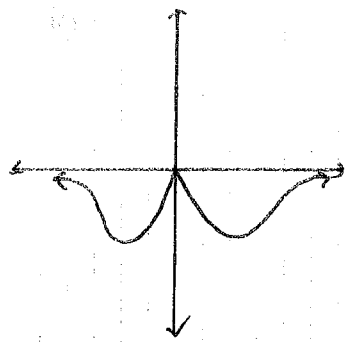
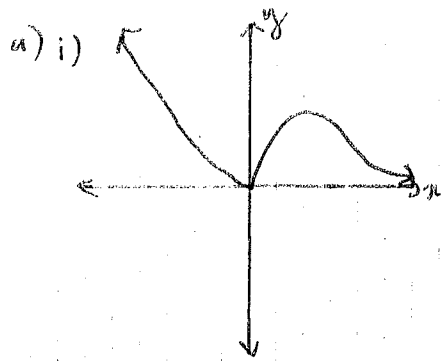
(c)

$$z \leq 2 \quad -\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{4}$$



(d)





b) $x^2 + xy + y^2 = 12$
 $2x + y + x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$

$$2x + y + \frac{dy}{dx}(x + 2y) = 0$$

$$\frac{dy}{dx} = -\frac{2x + y}{x + 2y}$$

When tangent is horizontal

$$\frac{dy}{dx} = 0$$

$$0 = -2x - y$$

$$y = -2x \quad (1)$$

$$x^2 + xy + y^2 = 12 \quad (2)$$

sub (1) into (2)

$$x^2 + x(-2x) + (-2x)^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2$$

When $x = 2$, $y = -4$

$x = -2$, $y = 4$

$\therefore$ tangent horiz at $(2, -4)$ $(-2, 4)$

c) i) $2x^3 - 3x^2 + 4x - 1$

$$2\left(\frac{x}{2}\right)^3 - 3\left(\frac{x}{2}\right)^2 + 4\left(\frac{x}{2}\right) - 1 = 0$$

$$x^3 - 3x^2 + 8x - 4 = 0$$

ii) $2(\sqrt{x})^3 - 3(\sqrt{x})^2 + 4(\sqrt{x}) - 1 = 0$

$$2\sqrt{x}(x+2) = 3x+1$$

$$4x(x+2)^2 = (3x+1)^2$$

$$4x^3 + 7x^2 + 10x - 1 = 0$$

Question 4

a) $V_{\text{shell}} = \pi(R^2 - r^2)h$

$$= \pi \left[(6-x)^2 - (6-(x+2x))^2 \right] y$$

$$= \pi (12 - 2x - 2x) dx y$$

$$= 2\pi y (6-x) dx \quad (\text{as } dx^2 \approx 0)$$

$$V_{\text{solid}} = \int_0^3 2\pi (6-x) (6x + x^2 - x^3) dx$$

$$= 2\pi \int_0^3 x^4 - 7x^3 + 36x dx$$

$$= 2\pi \left[\frac{x^5}{5} - \frac{7x^4}{4} + 18x^2 \right]_0^3$$

$$= \frac{13.77\pi}{10} \text{ units}^3$$

$$b) i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

at (x_1, y_1)

$$\frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1}$$

$$\therefore y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$a^2 y y_1 + b^2 x x_1 = b^2 x_1^2 + a^2 y_1^2$$

$$\frac{y y_1}{b^2} + \frac{x x_1}{a^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

$$= 1 \quad (\text{since } (x_1, y_1) \text{ lies on } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1)$$

$$ii) \frac{x}{4} + \frac{3\sqrt{3}y}{8} = 1$$

$$\frac{x}{4} + \frac{\sqrt{3}y}{6} = 1$$

iii) By symmetry of the ellipse, tangent passes through $(-1, -\frac{\sqrt{3}}{2})$

$$-\frac{x}{4} - \frac{3\sqrt{3}y}{8} = 1$$

$$-\frac{x}{4} - \frac{\sqrt{3}y}{6} = 1$$

iv) Chord passes through the origin.

$$m = \frac{\frac{3\sqrt{3}}{2} + \frac{3\sqrt{3}}{2}}{2}$$

$$= \frac{3\sqrt{3}}{2}$$

$$y = \frac{3\sqrt{3}}{2} x$$

i) Horizontally:

$$N \sin \theta - F \cos \theta = \frac{mv^2}{r}$$

$$N \sin \theta = \frac{mv^2}{r} + F \cos \theta \quad (1)$$

Vertically:

$$F \sin \theta + N \cos \theta = mg$$

$$N \cos \theta = mg - F \sin \theta \quad (2)$$

$$ii) N \sin^2 \theta = \frac{mv^2}{r} \sin \theta + F \cos \theta \sin \theta \quad (3) \quad [(1) \times \sin \theta]$$

$$N \cos^2 \theta = mg \cos \theta - F \cos \theta \sin \theta \quad (4) \quad [(2) \times \cos \theta]$$

$$N = \frac{mv^2}{r} \sin \theta + mg \cos \theta \quad ((3) + (4))$$

$$= \frac{mv^2}{r} \sin \theta + \frac{mgr \cot \theta \sin \theta}{r}$$

$$= \frac{m(v^2 + g r \cot \theta) \sin \theta}{r}$$

$\therefore W\hat{B}Z A$ is concyclic
(interior $\angle$ of quadrilateral equals interior opposite $\angle$)
 $W\hat{B}A = W\hat{Z}A$ ($\angle$ s in same segment)

(ii) $X\hat{C}W = X\hat{B}W$ (given)
 $\therefore WXBC$ is concyclic
(WX subtends two equal $\angle$ s)
 $W\hat{B}C + W\hat{X}C = 180^\circ$ (opposite $\angle$ s of a cyclic quadrilateral)

(iii) $W\hat{Z}A = W\hat{X}C$ (interior $\angle$ of cyclic quadrilateral equals interior opposite $\angle$ s)
 $\therefore W\hat{B}C + W\hat{Z}A = 180^\circ$
 $\therefore W\hat{B}C + W\hat{B}A = 180^\circ$

(iv) (i) let $u = x^n$ $v' = e^x$
 $u' = nx^{n-1}$ $v = e^x$
 $I_n = [x^n e^x]_0^1 - n \int_0^1 x^{n-1} e^x dx$
 $= e^1 - n I_{n-1}$

(ii) $I_2 = e - 2I_1$

$$I_2 = e - 2I_1$$

$$I_1 = e - I_0$$

$$I_0 = \int_0^1 e^x dx$$

$$= [e^x]_0^1$$

$$= e - 1$$

$$I_1 = e - e + 1 = 1$$

$$I_2 = e - 2$$

$$I_3 = e - 3e + 6 = 6 - 2e$$

$$I_4 = e - 24 + 8e = 9e - 24$$

$$u = x/\sqrt{1-x^2} \Rightarrow \frac{x}{\sqrt{1-x^2}}$$

$$\therefore x' = \frac{1}{1-x^2}$$

$$A = x \times 2y$$

$$= x \times 2\sqrt{1-x^2}$$

$$(ii) V = \lim_{x \rightarrow 0} \sum_{n=0}^4 x \sqrt{6+4n^2} dx$$

$$= \int_0^4 x \sqrt{6+4n^2} dx$$

$$= -\frac{1}{8} \int_0^4 y^{\frac{1}{2}} dy \quad \text{let } y = 6+4n^2$$

$$= -\frac{1}{8} \left[\frac{2y^{\frac{3}{2}}}{\frac{3}{2}} \right]$$

$$= -\frac{1}{12} \left[y^{\frac{3}{2}} \right]_0^4$$

$$= -\frac{1}{12} \times 64^{\frac{3}{2}}$$

$$= -\frac{128}{3}$$

$$\frac{dy}{dx} = -8x$$

$$-dy = 8x dx$$

(h) (i) let $u = \cos x$

$$\frac{du}{dx} = -\sin x$$

$$-du = \sin x$$

$$\int_1^{-1} -\frac{du}{\sqrt{1-u^2}}$$

$$= [\tan^{-1} u]_{-1}^1$$

$$= \frac{\pi}{4} - -\frac{\pi}{4}$$

$$= \frac{\pi}{2}$$

(ii) $\frac{dx}{du} = -1$

$$-dx = du$$

$$\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$$

$$= - \int_{\pi}^0 \frac{(\pi-u) \sin(\pi-u)}{1+\cos^2(\pi-u)} du$$

$$= \int_0^{\pi} \frac{\pi \sin u - u \sin u}{1+\cos^2 u} du$$

$$= \int_0^{\pi} \frac{\pi \sin u}{1+\cos^2 u} du - \int_0^{\pi} \frac{u \sin u}{1+\cos^2 u} du$$

$$m_A = -\frac{\frac{5}{k}}{kt}$$

$$= -\frac{1}{k^2}$$

eqn of tangent

$$y - \frac{5}{k} = -\frac{1}{k^2}(x - kt)$$

$$k^2 y - kt = -x + kt$$

$$x + k^2 y = 2kt$$

(ii) when $y=0$, $x=2kt$

$$m_A = \left(\frac{kt+2kt}{2}, \frac{\frac{5}{k}+0}{2} \right)$$

$$= \left(\frac{3kt}{2}, \frac{5}{2k} \right)$$

$$x = \frac{3kt}{2} \quad y = \frac{5}{2k}$$

$$xy = \frac{3e^4}{4}$$

$$\therefore \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1+\cos^2 x} dx$$

$$= \frac{\pi}{2} \times \frac{\pi}{2}$$

$$= \frac{\pi^2}{4}$$

Question 7:

a) i. $y = \cot x$
 $= \tan\left(\frac{\pi}{2} - x\right)$
 $y' = -\sec^2\left(\frac{\pi}{2} - x\right)$
 $= -\operatorname{cosec}^2 x$

ii. At $A(x=a)$:

$$\sin a = \cot a$$

$$\sin a = \frac{\cos a}{\sin a}$$

$$\sin^2 a = \cos a \quad \text{--- (1)}$$

$$y = \sin x$$

$$y' = \cos x$$

$$\text{At } A(x=a): m_1 = \cos a$$

$$m_1 \times m_2 = \cos a \times -\operatorname{cosec}^2 a$$

$$= \cos a \times -\frac{1}{\sin^2 a}$$

$$= \cos a \times -\frac{1}{\cos a} \quad [\text{from (1)}]$$

$$= -1$$

$\therefore$ curves intersect at right angles at A

iii. From (1):

$$\sin^2 a = \cos a$$

$$1 - \cos^2 a = \cos a$$

$$\cos^2 a + \cos a - 1 = 0$$

$$\cos a = \frac{-1 \pm \sqrt{1 - 4(-1)}}{2}$$

$$= \frac{-1 \pm \sqrt{5}}{2}$$

$$\sin^2 a = \frac{-1 + \sqrt{5}}{2} \quad [\text{from (1)}]$$

$$\begin{aligned} \operatorname{cosec}^2 a &= \frac{2}{-1 + \sqrt{5}} \times \frac{-1 - \sqrt{5}}{-1 - \sqrt{5}} \\ &= \frac{2(-1 - \sqrt{5})}{1 - 5} \\ &= \frac{-2(1 + \sqrt{5})}{-4} \\ \operatorname{cosec}^2 a &= \frac{1 + \sqrt{5}}{2} \end{aligned}$$

b) i.

$$F = \frac{mgR^2}{x^2}$$

$$\frac{mv^2}{x} = \frac{mgR^2}{x^2}$$

$$v^2 = \frac{gR^2}{x}$$

$$= \frac{10 \times (6.4 \times 10^6)^2}{6.403 \times 10^6}$$

$$v \approx 8000 \text{ ms}^{-1}$$

ii. $v = r\omega$
 $v = x\omega$

$$8000 = (6.403 \times 10^6) \omega$$

$$\omega = 1.249 \times 10^{-3} \text{ rad/s}$$

$$T = \frac{2\pi}{\omega}$$

$$= 5030 \text{ s}$$

$$= 1 \text{ h } 24 \text{ min}$$

iii. $F = \frac{mgR^2}{x^2}$

$$= \frac{300 \times 10 \times (6.4 \times 10^6)^2}{(6.403 \times 10^6)^2}$$

$$\approx 2997 \text{ N}$$

$$\begin{aligned}
 \text{c) i. } y &= \sin^{-1} x - \sqrt{1-x^2} \\
 &= \sin^{-1} x - (1-x^2)^{\frac{1}{2}} \\
 y' &= \frac{1}{\sqrt{1-x^2}} - \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot -2x \\
 &= \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} \\
 &= \frac{1+x}{\sqrt{1-x^2}}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } \frac{1+x}{\sqrt{1-x^2}} &= \frac{1+x}{\sqrt{(1+x)(1-x)}} \\
 &= \sqrt{\frac{1+x}{1-x}} \\
 \int_0^a \sqrt{\frac{1+x}{1-x}} dx &= \left[\sin^{-1} x - \sqrt{1-x^2} \right]_0^a \\
 &= \left(\sin^{-1} a - \sqrt{1-a^2} \right) - \left(\sin^{-1} 0 - \sqrt{1-0^2} \right) \\
 &= \sin^{-1} a + 1 - \sqrt{1-a^2}
 \end{aligned}$$

Question 8:

$$\begin{aligned}
 \text{a) i. } \sin(A+B) - \sin(A-B) &= \sin A \cos B + \cos A \sin B - (\sin A \cos B - \cos A \sin B) \\
 &= 2 \cos A \sin B
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } LHS &= \sin \frac{\theta}{2} (1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) \\
 &= \sin \frac{\theta}{2} + 2 \cos \theta \sin \frac{\theta}{2} + 2 \cos 2\theta \sin \frac{\theta}{2} + 2 \cos 3\theta \sin \frac{\theta}{2} \\
 &= \sin \frac{\theta}{2} + \left\{ \sin \left(\theta + \frac{\theta}{2} \right) - \sin \left(\theta - \frac{\theta}{2} \right) \right\} + \left\{ \sin \left(2\theta + \frac{\theta}{2} \right) - \sin \left(2\theta - \frac{\theta}{2} \right) \right\} \\
 &\quad + \left\{ \sin \left(3\theta + \frac{\theta}{2} \right) - \sin \left(3\theta - \frac{\theta}{2} \right) \right\} \\
 &= \cancel{\sin \frac{\theta}{2}} + \cancel{\sin \frac{3\theta}{2}} - \cancel{\sin \frac{\theta}{2}} + \cancel{\sin \frac{5\theta}{2}} - \cancel{\sin \frac{3\theta}{2}} + \sin \frac{7\theta}{2} - \cancel{\sin \frac{5\theta}{2}} \\
 &= \sin \frac{7\theta}{2} \\
 &= RHS
 \end{aligned}$$

$$\text{iii. } \sin \frac{\theta}{2} (1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) = \sin \frac{7\theta}{2}$$

$$\text{When } \theta = \frac{2\pi}{7}:$$

$$\begin{aligned}
 RHS &= \sin \frac{7 \left(\frac{2\pi}{7} \right)}{2} \\
 &= \sin \pi \\
 &= 0
 \end{aligned}$$

$$\sin \frac{\theta}{2} (1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) = 0$$

$$\sin \frac{\theta}{2} = 0$$

$$\text{But when } \theta = \frac{2\pi}{7}:$$

$$\sin \frac{\pi}{7} \neq 0$$

$$\therefore 1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta = 0$$

$$\text{b) i. } f(x) = e^x \left(1 - \frac{x}{8} \right)^8$$

$$u = e^x$$

$$u' = e^x$$

$$v = \left(1 - \frac{x}{8} \right)^8$$

$$v' = 8 \left(1 - \frac{x}{8} \right)^7 \cdot \left(-\frac{1}{8} \right)$$

$$= - \left(1 - \frac{x}{8} \right)^7$$

$$\begin{aligned}
 f'(x) &= -e^x \left(1 - \frac{x}{8} \right)^7 + e^x \left(1 - \frac{x}{8} \right)^8 \\
 &= -e^x \left(1 - \frac{x}{8} \right)^7 \left[1 - \left(1 - \frac{x}{8} \right) \right] \\
 &= -e^x \left(1 - \frac{x}{8} \right)^7 \left(\frac{x}{8} \right) \\
 &= \frac{-xe^x}{8} \left(1 - \frac{x}{8} \right)^7
 \end{aligned}$$

Stat points occur when $f'(x) = 0$:

$$\frac{-xe^x}{8} = 0$$

$$\left. \begin{array}{l} x=0 \\ y=1 \end{array} \right\}$$

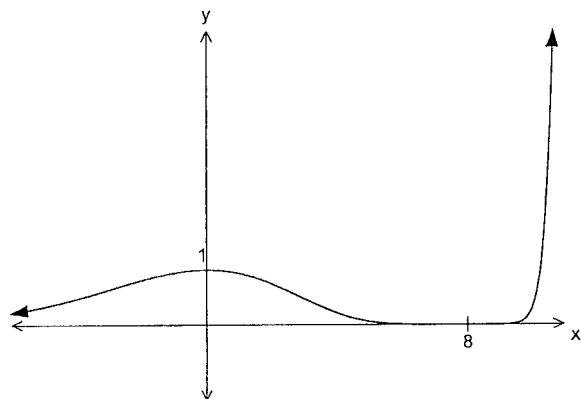
$$\left(1 - \frac{x}{8}\right)^7 = 0$$

$$1 - \frac{x}{8} = 0$$

$$\left. \begin{array}{l} x=8 \\ y=0 \end{array} \right\}$$

Stat. points at $(0,1)$ and $(8,0)$

ii.



iii. When $x < 8$:

$$e^x \left(1 - \frac{x}{8}\right)^8 \leq 1 \text{ from graph}$$

$$e^x \leq \frac{1}{\left(1 - \frac{x}{8}\right)^8} \text{ Note: } \left(1 - \frac{x}{8}\right)^8 > 0 \text{ when } x < 8$$

$$e^x \leq \left(1 - \frac{x}{8}\right)^{-8}$$

iv. When $x = 1$:

$$e \leq \left(1 - \frac{1}{8}\right)^{-8}$$

$$e \leq \left(\frac{7}{8}\right)^{-8}$$

$$e \leq \left(\frac{8}{7}\right)^8$$

When $x = -1$:

$$e^{-1} \leq \left(1 + \frac{1}{8}\right)^{-8}$$

$$\frac{1}{e} \leq \left(\frac{9}{8}\right)^{-8}$$

$$\frac{1}{e} \leq \left(\frac{8}{9}\right)^8$$

$$e \geq \left(\frac{9}{8}\right)^8$$

$$\therefore \left(\frac{9}{8}\right)^8 \leq e \leq \left(\frac{8}{7}\right)^8$$